Illinois State Water Survey
COOPERATING TECHNICAL PARTNERS (CTP)
FLOOD RISK PROJECT
MAPPING ACTIVITY STATEMENT (MAS)

MAS No. ISWS 20-09 Massac County Data Development

The Flood Risk Project described in this MAS dated June 2020 shall be completed in accordance with the CTP Partnership Agreement dated September 9, 2013 between Illinois State Water Survey (herein referred to as "CTP") and the Federal Emergency Management Agency (FEMA).

DISCLAIMER (do not delete): In recognition of the MIP Studies Redesign which was deployed in June 2017, FEMA Region V has made and effort to update the accompanying "R5_MAS_SOW_Workbook" to align with terminology, projects, purchases and tasks in the redesigned MIP. The Region acknowledges that the signed SOW may or may not contain the required sections, tables, scope or terminology that is fully aligned with the FEMA Region V Workbook. Due to the reasons, the "R5_MAS_SOW_Workbook" will be treated as the sole source for creating and obligating projects, purchases and tasks in the redesigned MIP.

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SECTION 1—OBJECTIVE AND SCOPE

The objective of the Flood Risk Project documented in this Mapping Activity Statement is to develop and/or support flood hazard data and program-related tasks through completing technical risk analysis and mapping activities. These activities may or may not result in a new or updated Flood Insurance Rate Map (FIRM) and Flood Insurance Study (FIS) report for one or more communities within the project area.

The objective of this study is to support (under separate future funding) a countywide DFIRM for Massac County and incorporated areas. Data Development tasks associated with this MAS will end at KDP-2.

Discovery within this watershed and county have not been performed or funded, nonetheless, hydrologic and hydraulic studies are being undertaken to update all unverified Zone A and Zone AE miles and re-delineate Ohio River miles within Massac County.

Data Development tasks within Massac County are within the Lower Ohio HUC8 (05140206) and include a combination of updates to effective Zone AE and Zone A Special Flood Hazard Areas, as well as new Zone A analysis and re-delineation the Ohio River. The IDNR-OWR has requested for Massac Creek (11.3 miles Zone AE) near Metropolis to be updated as part of this MAS.

The Brookport Levee System protects the Village of Brookport from Ohio River floods and other tributaries. Redelineation of the Ohio River and hydraulic analysis of Sevenmile Creek may be influenced by the effects of the levee, however, specific levee modeling and/or outreach associated with LAMP is not included in this MAS. FEMA and the RSC (STARR II) will be responsible for levee-related modeling and outreach, if needed.

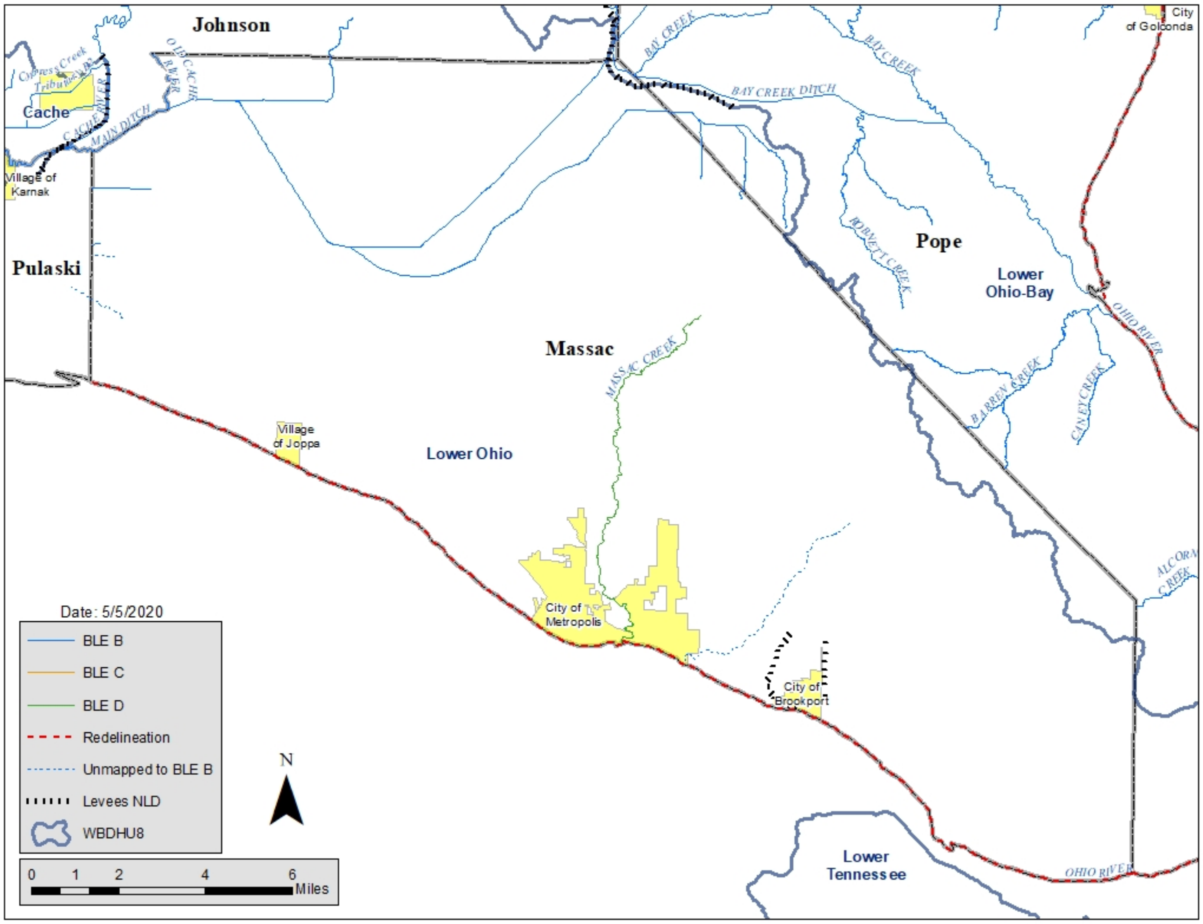
The Reevesville Levee System influences a significant amount of the hydrology and hydraulics within Massac County, as it prevents Ohio River flooding from connecting to Main Ditch from Bay Creek in Pope County. The Main Ditch network of streams serves as the historical flood path for the Ohio River overflow, prior to the building of the Reevesville Levee system in the early 1950's

The National Levee Database (NDL) lists a number of levees located in Massac County. These present a risk to the project as requirements for levee outreach and study may have evolved in the years since the current DFIRM became effective. ISWS will not be performing any levee specific outreach but will rely on FEMA and the RSC to provide support to resolve any issues that arise.

Main Ditch was most recently evaluated in 2014 by ISWS, in conjunction with the Cache River analysis in Pulaski County (ISWS MAS11-06 Pulaski County). This analysis was used to inform the Cache River analysis and was performed using unsteady-state 1D HEC-RAS but needs to be re-evaluated using the most recent topographic information and 2D HEC-RAS because of the braided and interconnected ditches and streams that form this system, and because of the complicated effects of the Cache River system that it is connected to and influenced by.

The 2D analysis of the Cache River Valley (including Main Ditch) will be performed as a regional analysis and will include portions of Alexander County, Pulaski County, Johnson County, Massac County, and Pope County. Each of these counties will reflect their appropriate portion of the regional analysis in their respective FFY20 MAS, including potential modification of ISWS MAS18-05 Alexander/Pulaski.

Figure 1. Massac County Data Development Streams and Study Types



Ohio Redelineation

No new Ohio River hydrologic or hydraulic studies are associated with this MAS. The effective hydrologic analysis is from the U.S. Army Corps of Engineers Ohio River Basin, Comprehensive Survey, Appendix C, "Hydrology" dated August 1966. The effective hydraulic analysis is from the U.S. Army Corps of Engineers, Ohio River Division, Ohio River Profiles Study dated January 1981. However, duplicate effective HEC-RAS models prepared by AMEC Earth & Environmental to the effective Ohio River HEC-2 models will be used for this project to support Ohio River floodplain re-delineation and development of a FEMA FIS flood profile. The HEC-RAS duplicate effective models, Ohio River Smithland Pool (Mile 918 to Mile 849) and Below Smithland Pool (Mile 981 to Mile 918), completed December 13, 2008 are a numerical representation of the previous study incorporating updated computations and software.

Elevations in the Comprehensive Survey, the Water-surface Profiles Study and HEC-2 models, and the HEC-RAS models are reported in the Ohio River Datum. IDNR-OWR completed a review of the HEC-RAS models in 2009 and found no reason to preclude their use for map modernization or in performing conversions between survey datums. The HEC-RAS water surface elevations will be converted from the Ohio River Datum to NGVD 1929 using methodology provided in the USACE Louisville District report Difference Between “1929 General Adjustment” and “Ohio River” Datum Planes dated November 29,

2005. Conversion from NGVD 1929 to NAVD 1988 will be accomplished using cross section by cross section conversion methodology. This method is applied to all Illinois reaches of the Ohio River for consistency within the state.

The Kentucky Division of Water has completed digital countywide regulatory products for its Ohio River counties. Therefore, Ohio River georeferenced cross sections will be obtained from the FEMA National Flood Hazard Layer database.

This project is proceeding under the collective assumption that the USACE 1981 Ohio River Profiles Study remains an appropriate representation of Ohio River water surface elevations for multiple events given completion of the Olmsted Lock and Dam (near RM 964.4, in Olmsted, IL) and the associated removal of Locks and Dam 52 (near RM 939, about 1.5 miles downstream of Brookport, IL) and Locks and Dam 53 (near RM 962, about 11 miles upstream of Cairo, IL).

The Ohio River is stationed upstream to downstream in miles below headwaters in Pittsburgh, PA. The following figure was produced by ISWS for MAS purposes, outside agency review has not been solicited.

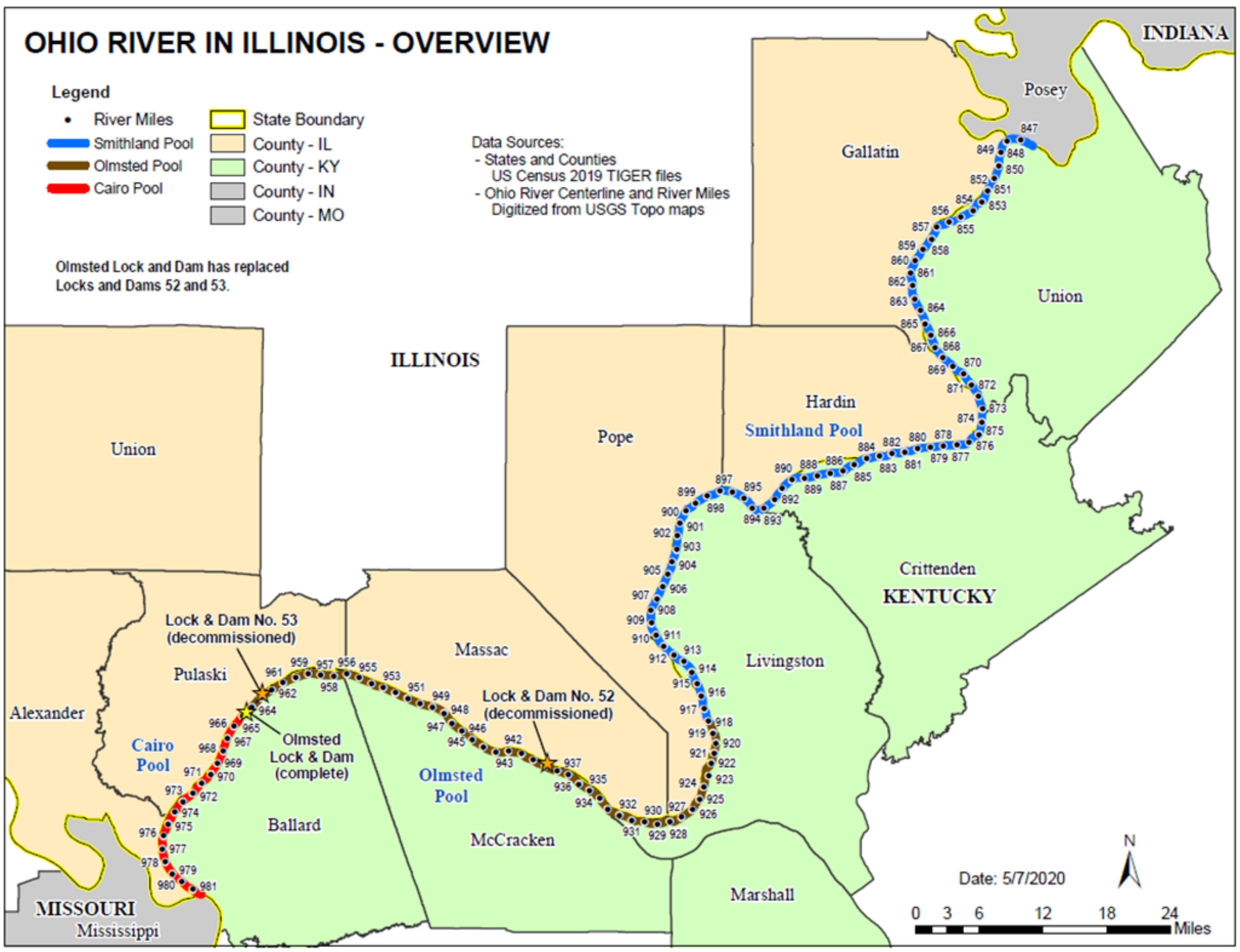


Figure 2. Ohio River Redelineation

The watersheds and jurisdictions in which Flood Risk Projects will be performed, as well as their applicable project type/activities, are summarized in Table 1.1: Flood Risk Project Watersheds and Jurisdictions. All applicable project activities should be identified in the last column of the table for each watershed, county/parish or community listed in Table 1.1 (e.g., Discovery only, Discovery and Flood Risk Products, FIRM/FIS updates, Prelim Distribution, Post-Preliminary Processing). Watershed Reports will be created and distributed to counties/parishes and communities identified as including Discovery in the Project Type.

Table 1.1: Flood Risk Project Watersheds and Jurisdictions

Watershed	Geographic Footprint	Counties/Parishes and Communities Included in Project	Project Type
Lower Ohio HUC8 (HUC05140206)	Massac County	Massac County City of Metropolis Village of Joppa City of Brookport	Data Development

Additionally, the CTP involved in this project will develop new and/or updated flood hazard data as summarized in Table 1.2: Total Stream or Shoreline Mile Counts by Type of Study. The FIRM and FIS report for the watersheds and areas identified in Table 1.2 will be produced in the North American Vertical Datum of 1988 (NAVD88).

Table 1.2: Total Stream or Shoreline Mile Counts by Type of Study

Type of Study*	Watershed or Jurisdiction	Riverine	Hydraulic Analysis Option (A-E)	1D or 2D
Updated Effective Zone VE and AE Studies	Massac County	New AE = 11.3 miles (Massac Creek) Leverage AE = 0 miles Re-Delineation AE = 28.2 miles (Ohio River)	BLE-D	1D
Updated Effective Zone A Studies	Massac County	New A = 27 miles (Main Ditch), 6 miles (tributaries) Leverage A = 0 miles	BLE-B	Main Ditch = 2D Tributaries = 1D and 2D (as needed)
New Zone A Studies Identified	Massac County	New A = 8 miles Leverage A = 0 miles	BLE-B	1D and 2D (as needed)

To complement Table 1.2, the CTP shall also deliver a GIS shapefile indicating stream lines and study method (with reach ID's as indicated in CNMS) for all mapping activities in this MAS.

This Flood Risk Project will be completed by the following entities:

- Illinois State Water Survey.

The CTP shall notify FEMA and all applicable parties of all meetings with community officials, and other relevant meetings, between three and six weeks prior to the meeting (with as much notice as possible). FEMA and/or its contractor may or may not attend the community meetings.

The CTP shall maintain an archive of all data submitted. All supporting data must be retained for three years from the date a funding recipient submits its final expenditure report to FEMA. The CTP must demonstrate to FEMA compliance with Subpart 24.1 of the Federal Acquisition Regulation (FAR) related to the handling of Personally Identifiable Information (PII) associated with the Activities listed in this MAS.

The CTP is responsible for providing a Quality Management Plan (QMP) to include an independent Quality Assurance/Quality Control (QA/QC) plan for all assigned activities. The CTP will submit a Summary Report that describes and provides the results of all automated or manual QA/QC review steps. The report should include the process for all assigned activities. The QMP is delivered directly to the FEMA Regions.

Independent QC review activities may be performed by the CTP or FEMA's contractor at the discretion of FEMA. If the CTP will be responsible for the QC review, the entity that will perform QC should be identified in this MAS. The CTP will need to submit its QC plan to the Regional Project Officer for approval.

Please note FEMA will also be performing periodic audits and overall study/project management to promote quality, including National Quality Reviews (QR) required per FEMA standards for all Flood Risk Projects. Whether or not the CTP performs the Independent QC review mentioned in the preceding paragraph, the CTP will be responsible for addressing any and all comments resulting from National QRs and any additional independent QA reviews required by the FEMA Regional Office, including re-submittal of deliverables as needed to pass technical or quality review. The CTP will submit regulatory products to FEMA's designated National QR reviewer (for QR 1-8 reviews) for review and approval prior to public issuance.

Metadata is required for certain activities. The FEMA current, applicable metadata guidelines will be followed.

FIRM-related tasks require a passing QC Report from FEMA's National FIRM database auto-validation tool, the Database Validation Tool (DVT) for QRs #1, #2, and #5 as required in FEMA standards. DVT can be found in the Mapping Information Platform tasks that correspond to the QRs: Draft FIRM Database Capture, Produce Preliminary Products Data Capture, and Final Mapping Products Data Capture. Training materials for this step are available on the Mapping Information Platform (MIP) at MIP User Care Training Materials.

FEMA will provide download/upload capability for data submittals through the MIP located at https://hazards.fema.gov. As each activity is completed, the data must be submitted to the MIP by the CTP or other designee. After data is submitted, a Validation task must be completed. Note that FEMA's designee will be assigned this task, not the CTP

The CTP assigned to the activity will respond to any comments generated as a result of the mandatory quality control checks by the Production and Technical Services (PTS) contractor. The PTS QC process is nationally funded and required on each non-PTS study.

In cooperation with the FEMA Project Officer, a Project Management Team (PMT) will be established by the CTP consisting of representatives from the CTP, FEMA's Regional Engineer, and other appropriate parties (e.g., FEMA contractors) at the discretion of FEMA. The PMT will be responsible for coordinating the activities identified in this MAS. The FEMA Region will be provided with documentation identifying the established PMT. The PMT will include representatives from the Illinois Department of Natural Resources/ Office of Water Resources, the FEMA Region 5 Engineer assigned to Illinois, and the Illinois State Water Survey (ISWS), Coordinated Hazard Assessment Mapping Program Section Head or their designee.

Responsible Mapping Partner: ISWS

Scope: Project Management is the active process of planning, organizing, and managing resources toward the successful accomplishment of predefined project goals and objectives. The CTP will coordinate with the FEMA Regional Office with respect to Project Management activities and technical mapping activities.

Earned Value Data Entry

The FEMA Mapping Information Platform (MIP) is designed to track the earned value of Flood Risk Projects. This information is automatically calculated by the MIP, using the actual cost and schedule of work performed, or "actuals," and comparing them to the expected cost and schedule of work performed, or "baseline."

Once the FEMA Regional Office has funded a project, FEMA or its contractor will create the tasks in the MIP. This step establishes the baseline for the project in the MIP, using the cost and schedule information for each task as outlined in this document.

The MIP study application allows FEMA and the CTP to manage the status of these projects at a task level. The cost and schedule information, updated monthly by the CTP for each contracted task, is compared to the baseline established for those tasks. This information is rolled up to a project level and monitored by the FEMA Region to assess progress and earned value.

Earned Value data entry involves updating cost, schedule, and performance (physical percent complete), and as of date in the MIP by the CTP each month for each assigned task.

Once the baseline has been established in the MIP, the CTP shall input the performance and actual cost to date for all tasks within each project for which the CTP is responsible. This must be completed at a minimum once every 30 days and at the completion of the task. When a task is completed and has been validated (if applicable), the CTP shall enter 100 percent complete, enter the actual completion cost, and the actual completion date within the Track Task Progress Workbench as applicable. The tasks will be available on the Track Progress workbench up to 90 days after the completion of the producer task in each purchase.

The MIP shall also be populated with appropriate leverage information regarding who (CTP or community) paid for the data provided and the amount of data used by the Flood Risk Project. The CTP will maintain a Schedule Performance Index (SPI) and Cost Performance Index (CPI) between 0.92 and 1.08. Special Problem Reports (SPR) explaining any variance must be submitted in a timely manner as required.

The CTP is required to report on the earned value of projects that are in the MIP monthly and must give explanations for variances outside of the tolerance defined above. FEMA Regional Offices must implement a Corrective Action Plan (CAP) when a CTP partner is outside of the tolerance. A CAP must define the reason for the variance and the intended resolution. FEMA Regional Offices must coordinate with FEMA Headquarters when CAPs are developed.

Program Management and overarching Community Outreach and Mitigation Strategies (COMS) SOW tasks are now tracked in the MIP. Cost and schedule performance measures must be defined and documented in those separate MAS or SOW. These measures must be used to monitor partner performance and to determine future funding eligibility. This exception only applies to tasks not able to be conducted or tracked in the MIP.

The Project Officer, as needed, may request additional information on status of the project on an ad hoc basis.

Standards: All PM work shall be performed in accordance with the standards specified in Section 5 – Standards.

Deliverables: The CTP shall deliver the following checked item(s) to the FEMA Regional Project Officer (please click/check the box of the deliverables included in this MAS):

☒	Monthly earned value data reporting through the MIP with variance explanations to support management of technical mapping activities within specified time frame, for both Regulatory and Flood Risk Products
☐	Management of SPI/CPI performance for an organization
☐	Overall project Quality Management Plan including QA/QC maintenance information, such as maintaining a QA/QC log and providing a QA/QC approach to FEMA for review and approval
☐	Management of adherence to scope of work and quality of work for an organization
☐	Other: **{Insert additional details}**.

Project Risk Identification and Mitigation

Responsible Mapping Partner: ISWS.

Risks to the planned completion of a project may come from various sources. Risks should be identified during the planning phase and monitored throughout the project, so that potential impact can be assessed, and solution strategies developed and implemented as needed.

Table 1.3: Project Risk Identification

Project Risk	Potential Impact	Solution Strategy
Unidentified Levee Systems, including agricultural levees, not shown on NLD or NFHL	Schedule and funding impacts resulting from necessary outreach, including supporting modeling and mapping efforts.	Initiate coordination discussion with STARR II at beginning of project/grant. STARR II will oversee levee communication and outreach with stakeholders. Levee analysis is not funded with this MAS.
Reevesville Levee System	Schedule and funding impacts for: hydrology and hydraulics of data development tasks along Main Ditch and tributaries.	Per FEMA's direction, the Reevesville Levee system will be assumed to be accredited while FEMA continues to coordinate with USACE regarding accreditation requirements.
Field Survey	Schedule delay from seasonal conditions and/or circumstances outside of survey contractor control.	Initiate field survey contracts at beginning of project/grant.
Covid-19	Schedule delay and funding impacts from Covid-19 related illness and/or directives/policies by local, state, and federal entities. In-person meetings may be impacted and require the use of alternatives such as Zoom or other video and/or phone-based technology instead.	Follow all local, state, and federal directives/policies and assess project impacts on a recurring schedule.
No Discovery	Discovery was not performed prior to this data development MAS, therefore it is more possible that local issues and needs will be identified that	ISWS will initiate communication with local officials and stakeholders at the beginning of the project/grant. The Project Initiation meeting or coordination call will be the first official outreach task by

Project Risk	Potential Impact	Solution Strategy
	require additional time and/or cost to address.	ISWS. ISWS will coordinate with FEMA to address needs and feedback initiated by local communities.
Ohio River Re-Delineation	Schedule delay and funding impacts associated with potential changes to Ohio River pool elevations associated with updates to the Ohio River model by others.	ISWS will coordinate with FEMA to ensure that the most up-to-date information from USACE is accessible to ISWS/

Event Data Capture

> *NOTE: The performance of community engagement and outreach takes place throughout the life of the Flood Risk Project. Work with your FEMA Region to develop a Project Outreach and Communication Plan.*

An alternate tracking method is acceptable with approval from the FEMA Regional Office. Up to 10 percent of the total budget can be used for Community Engagement. An additional (up to) 10 percent of the total budget can be used for Project Outreach. The activities identified here are for project specific community engagement and outreach. The overarching Community Outreach and Mitigation Strategies (COMS) Statement of Work (SOW) can be used for broader program wide engagement activities.

Responsible Mapping Partner: ISWS.

Scope:

Risk Communication & Outreach

Risk Communication & Outreach activities funded through this MAS shall not supplant or duplicate Community Engagement activities funded through other grants or contracts. There is a separate Statement of Work document used to document additional program wide COMS activities performed by CTPs.

The activities listed below are intended to support on-going project efforts that foster working relationships between federal, state, tribal and local governments with private and public interests to reduce the impact of natural disasters where a community's interaction with a natural hazard may be altered or averted. These activities should be reviewed for implementation throughout the Risk MAP lifecycle, including the defined meetings within it.

The following meetings and public outreach will be initiated by ISWS with project stakeholders, including local, state, and federal partners:

1) Project Initiation meeting, via conference call, and other project team meetings, as needed through the duration of the data development tasks described within this MAS. Discovery was not performed in this watershed; therefore, community input will be especially helpful in understanding local issues and locating available information.

2) Public news release to notify local stakeholders of field survey activities within their area, including direct communication via phone and/or email with local officials, and mailed notices to specific property owners whose land may require access by field survey personnel

3) In accordance with FEMA guidance, send SID-620 notification letters to project stakeholders for comment and to make them aware of the proposed study methods for the data development tasks described within this MAS

4) Flood Risk Review meeting with local officials and stakeholder organizations to review and discuss the draft results of data development tasks described in this MAS

5) Submittal of data development materials to IDNR-OWR for review and concurrence, as necessitated by state statutes, or as courtesy otherwise

6) In accordance with FEMA guidance, send SID-621 notification letters to project stakeholders to make them aware of available database products, including related work maps, completed models, and associated information for the data development tasks described within this MAS.

It is expected that including these approaches will assist in increased risk awareness and the development of actionable measures to reduce risk within the Flood Risk Project footprints. These activities should be implemented as directed by the Project Officer to:

- Increase the understanding of natural hazard risk within a community.
- Support local efforts to reduce natural hazard risk within a community or watershed area.
- Increase the effectiveness of meetings and engagement opportunities with communities throughout the Risk MAP life cycle.

30-Day Review of Proposed Models

Before any data development tasks supporting a Flood Risk Project that includes a FIRM update begin, each community affected by a new or updated regulatory products (FIS or FIRM) must be notified of the planned model or models to be used. The affected communities will be provided with a 30-day period beginning upon notification to consult with FEMA and the CTP regarding the appropriateness of the mapping model or models to be used. The results of the consultation do not necessarily guarantee that a change should be made, and decisions should be clearly documented. This consultation does not waive or otherwise affect the right of the community to later appeal any flood hazard determinations.

30-Day Review of Completed Models, Work Maps, and Database

When a Flood Risk Project will include new or updated FIRM panels, the CTP must provide access to the draft FIRM database and other contributing data, as requested, to the communities by the conclusion of Quality Review 1. The CTP also must provide the affected communities with a 30-day period during which the communities may provide data to FEMA and the CTP that can be used to supplement or modify the existing data. The CTP should incorporate any community-submitted data into the project as appropriate.

In order to be directly incorporated into the study, data or information submitted must be consistent with prevailing engineering principles or demonstrate scientific incorrectness by one or more of the following:

- Identifying and providing documentation of the methods or assumptions purported to be scientifically incorrect.
- Providing supporting data as to why the methods or assumptions used are not appropriate.
- Providing new or alternative analysis and mapping data utilizing methods consistent with prevailing engineering principles and meeting FEMA's Standards.
- Providing technical information or data indicating why the provided new or updated analysis and mapping should be accepted as more correct.

Television and Radio Outreach

The Project Team, in coordination with the appropriate staff in the FEMA Region Office of External Affairs, other FEMA staff, and community officials, shall engage with local radio and television outlets in an effort to further educate property owners about flood map revisions and appeals processes. FEMA cannot fund advertising, so public service announcements should be considered as a prime opportunity to meet this standard. Any engagement with the media should include explanations of the entire appeal process for flood hazard information, including comments on other information on the FIRM and Flood Insurance Study (FIS) report. Users should review the Stakeholder Engagement – Due Process Guidance document available at https://www.fema.gov/media-library/assets/documents/34953 and SID 622 https://www.fema.gov/media-library/assets/documents/35313 for more information.

Meetings

Flood Risk Projects will include in-person opportunities to engage communities, build risk awareness, increase capabilities for risk communication, and stimulate mitigation action at the local level. The overarching goal is to create a climate of understanding and ownership of the mapping process at the state and local levels. Well-planned and executed community engagement and project outreach can reduce political stress, confrontation in the media, and public controversy, which can arise from lack of information, misunderstanding, or misinformation. These engagement and outreach activities also can assist the CTP, FEMA, and other members of the PMT in responding to congressional inquiries. The MIP has the capability to capture all meeting information including sign in sheets, attendee lists, meeting minutes, and shared documents.

Provisions should be made for remote access video/audio feeds for those that cannot attend in person. These opportunities consist of:

Project Initiation Meeting/Coordination Call. This meeting will serve to discuss the project scope and timeline, set expectations for communities with regard to risk communication, share methods and data to be used in the mapping efforts if included in the project, and answer any local questions about the project. If regulatory product updates are included, this meeting can also serve as the required coordination with communities regarding the expected results (increasing/decreasing flood hazard areas/depths, etc.). This can be held as an in-person meeting, webinar, or conference call. *Note: although not required unless regulatory products are involved, this call/meeting is especially helpful for introducing the project to the involved communities when six months or more has passed since the post-Discovery coordination.*

Flood Risk Review Meeting. This meeting will serve to provide communities with engineering data and drafts of Flood Risk products, collect feedback, and revise as needed. It will also provide the opportunity to show how the datasets and outreach tools can help communities become more resilient by understanding risk data, communicating about risk, prioritizing mitigation actions and improving mitigation plans, especially risk assessments and mitigation strategies. Activities include planning, presenting, and facilitating discussions of data inputs and engineering models used for flood studies with community officials. In addition, draft work maps showing initial study results will be presented during the meeting. This meeting may also be used to kick off or occur during the 30-day review of materials at the workmap stage (workmap, associated models, etc.) required. Note: It is also recommended that for a coastal storm surge study a meeting be held with the community and additional stakeholders as appropriate to review the results of the storm surge modeling before preceding to modeling coastal wave heights and mapping.

Activities may include:

- Develop an engagement plan to reach the key community stakeholders, such as local officials and community partners, during the Flood Risk Review period.
- Develop or enhance relationships with key community stakeholders to increase the reach of messages about risk and to improve the local will and ability to take mitigation actions.
- Drive attendance through outreach to meeting invitees through personalized follow-up emails and calls.
- Conduct webinar to further engage and inform attendees.
- Track engagement efforts and responses to reflect ongoing work.
- Facilitate meeting.
- Talk with local officials regarding their desired role in engaging the public after the issuance of Preliminary FIRMs.
- Develop/update an outreach toolkit (select from materials such as fact sheets, talking points, FAQs, and brochures).
- Revise or create a Community Profile and/or dashboard to reflect the most current community information and insights.
- Conduct a post-meeting review session with the project team and provide a future recommendations report.
- Identify and document community commitments, such as mitigation action and engagement activities, to inform future interactions with the community. Update and collect Project Charters (when applicable).

Status Reports

In addition to Risk MAP meetings, to facilitate information sharing and a continuing dialogue between the PMT and the community, the CTP will provide communities with regular status reports outlining the current project status, key accomplishments to date, identified risks, if any, and next steps including estimated next meeting date and meeting content (template to be provided from FEMA or can be created by Mapping Partner). These status reports will be provided to FEMA for review before electronic distribution. Project update status reports will be distributed to communities at mid-points between each of the meetings, and between the Final Meeting and effective date (if included in this MAS), to help introduce and prepare the communities for upcoming discussions.

Project Outreach and Communication Plan (POCP)

The CTP will work with the FEMA Regional Office during the initiation of this activity to develop the Project Outreach and Communication Plan (POCP) to support the implementation of the mapping project. The FEMA Regional Office will have access to many customizable outreach tools that have been developed for this process to support each touchpoint that the PMT has with the community.

Standards: All Community Engagement and Outreach work shall be performed in accordance with the standards specified in Section 5 – Standards. All communication with local governments will be done in accordance with 44 CFR Part 66.

Deliverables: The CTP shall deliver the following checked items to the FEMA Regional Project Officer in accordance with the schedule outlined in Section 6 – Schedule and include within the Technical Support Data Notebook (TSDN):

☒	A Project Communication Plan detailing outreach and coordination activities. NOTE: The MIP has tasks available to capture Outreach activities. Upon completion of these tasks, please upload relevant data to those tasks in the MIP.[1]
☐	Watershed/Community Assessment outputs, including logs of telephone discussions (if applicable)
☒	Meeting invitation, agenda, presentation slides (as requested), and meeting notes for FEMA review
☐	Action Identification and Advancement Plan (if applicable)
☐	Project update status reports for project communities
☒	Provide documentation of adherence with the requirements for the community 30-day review of proposed models and 30-day review of work maps, completed models, and associated information

[1] All products should be uploaded to the MIP (or submitted to MIP Help on a hard drive if they are over the size listed in the MIP Guidance document)

Survey Data Capture

Note: CTPs should consider including and completing this task in the MIP if the partner believes data will need to be submitted during the course of the project, even if conducting field survey is not included in the MAS. In addition, leveraged field survey data must be documented and submitted for this task. Failure to submit this data could result in the workflow being reverted to production tasks and could lead to a delay in the schedule. If the CTP obtains field survey data that was not included in the MAS, the CTP shall contact the Region to submit a change request to include the task and leverage as part of the award and request that this task be added to the study in the MIP.

Responsible Mapping Partner: ISWS.

Scope: To supplement any field reconnaissance conducted during the Discovery phase of this project, the CTP shall conduct a detailed field or desktop reconnaissance of the specific study area to determine conditions along the floodplain(s), types and numbers of hydraulic and/or flood-control structures, apparent maintenance or lack thereof of existing hydraulic structures, locations of cross sections to be surveyed, and other parameters needed for the hydrologic and hydraulic analyses.

The CTP shall conduct field surveys, including obtaining channel and floodplain cross sections, identifying or establishing temporary or permanent benchmarks, and obtaining the physical dimensions of hydraulic and flood-control structures. If appropriate, the CTP shall also identify items needed for coastal analyses including land cover, vegetation types, housing, dunes, beach nourishment, coastal structures, and transects. The CTP shall also coordinate with other entities that are involved in the Topographic Data Development process regarding ongoing activities and deliverables.

Existing survey data, or as-built data, may be used to produce flood studies and related products. However, if existing data are to be collected, the FEMA Region should be consulted before leveraging the best available existing survey data to ensure acceptability for the intended level of flood hazard study.

Standards: All Survey work shall be performed in accordance with the standards specified in Section 5 – Standards.

Deliverables: The CTP shall produce items listed in the Survey Data Capture section within the most currently dated "Technical Reference: Data Capture" document, located at https://www.fema.gov/media-library/assets/documents/34519, made available to FEMA by uploading the digital data to the MIP in accordance with the schedule outlined in Section 6. Where paper documentation is required by state law for professional certifications, the CTP may submit the paper in addition to a scanned version of the paper for the digital record. Please coordinate with the regional and/or state representative to verify state reporting requirements.

Existing or New Topographic Data Capture and Terrain Data Capture

Note: Every MAS for a flood data update must include this task to ensure the topography used is documented and submitted. In addition, leveraged topographic data must be documented and submitted for this task. For coastal studies, include collection or leveraged bathymetry data plans.

Responsible Mapping Partner: ISWS.

Scope: Topographic/elevation data may be new or existing. New is defined as data that will be flown and processed for the areas specified in this MAS as study areas according to the referenced specifications. Existing topographic/elevation data (previously flown and/or processed) may be used to produce flood studies and related products. However, if new data is not to be collected, the FEMA Region should be consulted before leveraging the best available existing topographic to ensure acceptability for the intended level of flood hazard study.

The CTP shall obtain topographic data for the floodplain areas to be studied including overbank areas. These data will be used for hydrologic analysis, hydraulic analysis, floodplain boundary delineation and/or testing of floodplain boundary standard compliance. The CTP shall gather availability, currency, and accuracy information for existing topographic data covering the affected communities in this MAS. The CTP shall use topographic data for work in this MAS only if it is better quality than that of the original study or effective studies. The Mapping Partner will ensure that the FEMA Geospatial Data Coordination Policy and Implementation Guide is followed, and the data obtained or to be produced are documented properly as per those policies and guidelines.

Requirements for Existing Topographic Data Capture: The CTP shall use topographic data for the areas described in the Table 1.5 Summary of Topographic Data table. The source of the topographic data must be listed as well. The CTP shall coordinate with other team members conducting field surveys as part of this MAS. The CTP should follow the guidelines set forth in the Data Capture Standards Guidance, Technical Reference and the Elevation Guidance documentation to determine the suitability of the existing dataset in terms of currency and accuracy. The CTP should confirm with the FEMA Project Officer the use of leveraged topographic data.

In addition, the CTP shall address all concerns or questions regarding the topographic data development that are raised during the Independent QA/QC review.

Requirements for development of Terrain Data Capture: For this activity, the CTP shall utilize the data collected under the New and/or Existing Topographic Data Capture task and via field surveys to create a best available digital elevation model for the subject flooding sources. The CTP should confirm with the FEMA Project Officer the appropriate data model(s) (i.e. DEMS) for the intended use of the data.

In addition, the CTP shall address all concerns or questions regarding the topographic data development that are raised during the Independent QA/QC review.

Table 1.5: Summary of Topographic Elevation Data

Watershed/ Flooding Source	New OR Existing	Year Acquired	Data Type	Vertical Accuracy (RMSEz)	Source/Data Vendor (e.g. Public domain, community supplied, procured as part of this MAS, etc.)	Contact Information	Use Restrictions
Lower Ohio HUC8 tributaries within Massac County, IL (HUC05140206)	Existing	2012	LiDAR	FVA of 0.880 feet at the 95th percentile	Illinois Geospatial Data Clearinghouse	isgs@isgs.illinois.edu	none

Standards: All Topographic Elevation Data tasks shall be performed in accordance with the standards specified in Section 5 – Standards.

Deliverables: The CTP shall produce items listed in the New or Existing Topographic Data Capture section within the most currently dated "Technical Reference: Data Capture" document, located at https://www.fema.gov/media-library/assets/documents/34519, made available to FEMA by uploading the digital data to the MIP in accordance with the schedule outlined in Section 6. Where paper documentation is required by state law for professional certifications, the CTP may submit the paper in addition to a scanned version of the paper for the digital record. Please coordinate with the regional and/or state representative to verify state reporting requirements.

Base Map Data Capture

Note: Every MAS for a flood data update must include this task to ensure the base map used is documented and submitted. In addition, leveraged base map data must be documented and submitted for this task.

Responsible Mapping Partner: ISWS.

Scope: Base map preparation activities consist of obtaining and formatting the digital base map (raster or vector) for the project. Activities also include the review of the base map and obtaining the necessary documentation/verification that base map data source(s) may be used and distributed by FEMA. This task is equivalent to the Base Map Data Capture task in the MIP. The CTP shall prepare and provide the digital base map, including:

- Obtain digital files (raster or vector) of the base map. In coordination with the partner who performed Project Discovery, ensure that the FEMA Geospatial Data Coordination (GDC) Policy and Implementation Guide are followed.

- Secure necessary permissions from the base map source to allow FEMA's use and distribution of hardcopy and digital map products using the digital base map, free of charge.
- Review and supplement the content of the acquired base map to comply with FEMA standards.
- For the base map components that have a mandatory data structure, convert the base map data to the format required in FEMA standards.
- Determine that the digital data meets the minimum standards and specifications that FEMA requires for FIRM production.

Standards: All base map preparation work shall be performed in accordance with the standards specified in Section 5 – Standards.

Deliverables: The CTP shall produce items listed in the Base Map Data Capture section within the most currently dated "Technical Reference: Data Capture" document, located at https://www.fema.gov/media-library/assets/documents/34519, made available to FEMA by uploading the digital data to the MIP in accordance with the schedule outlined in Section 6.

Hydrology Data Capture

CTPs must complete this task in the MIP if an updated hydrologic analysis is completed. In addition, leveraged hydrology data must be documented and submitted to the MIP for this task. Failure to submit this task when data should have been submitted is no longer an issue. The MIP redesign allows for tasks to be reopened so that data can be uploaded. Note, however, if the CTP obtains hydrology data that was not included in the MAS, the CTP shall contact the region to submit a change request to include the task and leverage as part of the award and request that this task be added to the MIP workflow.

Responsible Mapping Partner: ISWS

Scope: The CTP shall perform hydrologic analyses for the flooding source(s) identified in Table 1.7: Summary of Hydrologic Analyses. Hydrologic analysis activities include the determination of peak flood discharges, the use of rainfall-runoff models, regression equations, gage analysis, and hydrograph development to support the level of detail required for the project. The CTP shall calculate peak flood discharges and/or flood hydrographs for the 10, 4, 2, 1, "1 plus" and 0.2 percent annual chance events using the analysis method listed in Table 1.7. These flood discharges will be the basis for subsequent Hydraulic Analyses performed under this MAS. In addition, the CTP will be responsible for addressing any and all comments resulting from the independent QC, including resubmittal of deliverables as needed to pass the technical review.

The CTP shall document automated data processing and modeling algorithms and provide the data to FEMA to ensure these are consistent with FEMA standards. Digital datasets (such as elevation, basin, or land use data) are to be documented and provided to FEMA for approval before performing the hydrologic analyses to ensure the datasets meet minimum requirements. If non-commercial (i.e., custom-developed) software is used for the analysis, then the CTP shall provide full user documentation, technical algorithm documentation, and the software to FEMA for review before performing the hydrologic analyses.

The CTP will compare the calculated, or computed, discharge with discharge determined from reliable gage data, if any. This comparison will only be done at locations where the two discharge values are considered representative of the same flooding source. Results of this comparison will be used in making a professional judgment for determining the discharge to be used for the hydraulic analysis.

Table 1.7: Summary of Hydrologic Analyses

Study Area/Flooding Source	Method	Square Miles of Leveraged Hydrology	Square Miles of New Hydrology
Lower Ohio HUC8 tributaries within Massac County, IL (HUC05140206)	USACE HEC-HMS (for Zone AE miles)	N/A	TBD
Lower Ohio HUC8 tributaries within Massac County, IL (HUC05140206)	USGS StreamStats (for 1D Zone A miles) USGS HEC-RAS Rain-on-Grid (for 2D Zone A miles)	N/A	TBD

Standards: All Hydrologic Analyses work shall be performed in accordance with the standards and guidance specified in Section 5 – Standards.

Deliverables: The CTP shall produce items listed in the Hydrology Data Capture section within the most currently dated "Technical Reference: Data Capture" document, located at https://www.fema.gov/media-library/assets/documents/34519, made available to FEMA by uploading the digital data to the MIP in accordance with the schedule outlined in Section 6. Where paper documentation is required by state law for professional certifications, the CTP may submit the paper in addition to a scanned version of the paper for the digital record. Please coordinate with the regional and/or state representative to verify state reporting requirements.

Independent QA/QC: Hydrology Data Capture

Responsible Mapping Partner: ISWS .

Scope: The Independent QA/QC Mapping Partner shall perform an impartial review of the technical, scientific, and other information submitted by the CTP specific to the hydrologic analyses to ensure that the data and modeling are consistent with FEMA standards and standard engineering practice and are sufficient to prepare the FIRM. FEMA may audit or assist in these activities if deemed to be necessary by the Regional Project Officer. This work shall include, at a minimum, the activities listed below. Please note FEMA will also be performing periodic audits and overall study/project management to ensure study quality. Where paper documentation is required by state law for professional certifications, the CTP may submit the paper in addition to a scanned version of the paper for the digital record. Please coordinate with the regional and/or state representative to verify state reporting requirements.

- Review the submittal for technical and regulatory adequacy, completeness of required information, and supporting data and documentation. The technical review is to focus on the following:
    - Use of acceptable models;
    - Use of appropriate methodology(ies);
    - Correctly applied methodology(ies)/model(s), including QC of input parameters;
    - Comparison with gage data and/or regression equations, if appropriate;
    - Comparison with discharges for contiguous reaches or flooding sources throughout the watershed.
- Verify that the data was submitted under the applicable folders on the MIP as described in the “Technical Reference: Data Capture” and “Guidance: Data Capture” documents.
- Maintain records of all contacts, reviews, recommendations, and actions and make the data readily available to FEMA.
- The reviewing Mapping Partner must document the results of the review in a memorandum or letter, send it to the Mapping Partner that performed the hydrologic analysis, and post it to the MIP through the Independent QA/QC of Hydrologic Analyses task. The review document must present specific comments and may include any new calculations or model runs in support of the review.

Standards: All Independent QA/QC work shall be performed in accordance with the standards specified in Section 5 – Standards.

Deliverables: The responsible Mapping Partner(s) shall produce items listed in the Hydrology Data Capture section within the most currently dated “Technical Reference: Data Capture” document, located at https://www.fema.gov/media-library/assets/documents/34519, made available to FEMA by uploading the digital data to the MIP in accordance with the schedule outlined in Section 6. Additionally, the TSDN must be delivered in accordance with Section 2 – Technical and Administrative Support Data Submittal.

This submittal will occur in accordance with the schedule outlined in Section 6 – Schedule.

- A Summary Report that documents the findings of the independent QA/QC review.
- Recommendations to resolve any problems that are identified during the independent QA/QC review.

Hydraulics Data Capture

CTPs must complete this task in the MIP if an updated hydraulic analysis is completed. In addition, leveraged hydraulic data must be documented and submitted to the MIP for this task. Failure to submit this task when data should have been submitted is no longer an issue. The MIP redesign allows for tasks to be reopened so that data can be uploaded. If the CTP obtains hydraulic data that was not included in the MAS, the CTP shall contact the region to discuss the potential need for a change request to include the task and leverage as part of the award and request that this task be added to the MIP workflow.

Responsible Mapping Partner: ISWS.

Scope: The CTP shall perform hydraulic analyses as described in Table 1.8: Summary of Hydraulic Analyses. Hydraulics analysis activities include establishing and reviewing regulatory floodways and flood elevations for the 10 -, 4 -, 2 -, 1 -, "1- plus" and 0.2 percent annual chance events based on flood discharge rates computed under Develop Hydrologic Data. The hydraulic methods used for this analysis may include base level and/or enhanced level hydraulic modeling. The base level will use an automated hydraulic model and use the best available elevation data to model the 10-, 4-, 2-, 1-, "1- plus" and 0.2 percent annual chance events. It may not include field surveys, floodways, or mapped Base Flood Elevations (BFEs) but will include mapped A or AE zones per Table 1.8. The enhanced level may include field surveys, floodways, and the 10-, 4-, 2-, 1-, 1- plus and 0.2 percent annual chance events, using methods described in Table 1.8. The Mapping Partner, at a minimum, must delineate the floodplain and floodway, if applicable, boundaries of the base flood. The Mapping Partner must also delineate the floodplain boundaries associated with the 0.2-percent-annual-chance flood, if it is calculated.

Exhibit 1.1 Hydraulic Analysis Options for Base and Enhanced Level Engineering

Option	Cross Sections	Flow Paths (Left, Right and Channel)	Manning's "n" Values	Structures	Flood Zone
A	Auto-placed; may be unnaturally straight with computerized look to them adjusted or auto-placed by "intelligent" methods.	Reach lengths are assumed equal.	Single value for each cross section.	Not included; cross sections placed as if structures don't exist or cross sections placed appropriately for structure modeling.	A
B	Auto-placed and hand adjusted or auto-placed by "intelligent" methods.	Reach lengths computed by offsetting stream centerline.	Overbanks from Land Use Land Cover (LULC) data, channel value estimated separately.	Not included; but cross sections placed appropriately for structure modeling.	A

Option	Cross Sections	Flow Paths (Left, Right and Channel)	Manning's "n" Values	Structures	Flood Zone
C	Each section reviewed by engineers.	Reach lengths adjusted based on draft floodplain.	Overbanks LULC data, channel value estimated separately.	Included; structure data from national, state or other data source. Estimated based on topography and aerial photos for those not available.	A
D	Each section reviewed by engineers.	Reach lengths adjusted based on draft floodplain.	Overbanks from LULC data, channel value estimated separately and calibrated where possible.	Included; structure data from as-builts, design plans, "measured" in the field, or other community datasets with opening information.	A or AE
E	Each section reviewed by engineers, channel bathymetry included in sections.	Reach lengths adjusted based on draft floodplain.	Overbanks from LULC data and field data, channel value estimated separately from field data and calibrated where possible.	Included; structure data from field survey, as-builts, design plans, "measured" in the field.	AE

The CTP shall use the cross-section and field data collected during Survey Data Capture and the topographic data collected during the Topographic Data Capture, when appropriate, to perform the hydraulic analyses. The hydraulic analyses will be used to establish flood water surface elevations, floodplain extents, and regulatory floodways for the listed study area or flooding sources.

If applicable, the CTP shall use the FEMA CHECK-2 or CHECK-RAS checking program to verify the reasonableness of the hydraulic analyses. To facilitate the independent QA/QC review, the CTP shall provide explanations for unresolved messages from the CHECK-2 or CHECK-RAS program, as appropriate. In addition, the CTP shall address all concerns or questions regarding the hydraulic analyses that are raised during the independent QA/QC review including resubmittal of deliverables as needed to pass the technical review.

The CTP shall document automated data processing and modeling algorithms for all GIS based studies and provide the data to FEMA for review to ensure these are consistent with the standards outlined above. If non-commercial (i.e., custom-developed) software is used for the analyses, then the CTP shall provide full user documentation, technical algorithm documentation, and software to FEMA for review before performing the hydraulic analyses.

Any flooding sources associated with a levee that are mapped as providing protection on effective FIRMs but will not meet certification requirements for the new FIRMs, will require revised hydraulic analysis. This revised analysis should be done in accordance with FEMA standards and guidance, as appropriate.

Table 1.8: Summary of Hydraulic Analyses

Study Area/Flooding Source	Hydraulic Analysis Option *	Total Miles of New Base Level or Enhanced Level Hydraulics	Description of Level of study- i.e., AE with BFE or A zone mapping
Massac Creek	BLE-D	11.3 miles	Zone AE
Ohio River Tributaries	BLE-B	8.2 miles	Zone A (1D)
Cache River Tributaries	BLE-B	27 miles (Main Ditch 2D) 6 miles (tributaries 1D)	Zone A (2D) Zone A (1D)

*List Option letter from Exhibit 1.8 listed above.

Standards: All Hydraulic Data work shall be performed in accordance with the standards specified in Section 5 – Standards.

Deliverables: The CTP shall produce items listed in the Hydraulics Data Capture section within the most currently dated "Technical Reference: Data Capture" document, located at https://www.fema.gov/media-library/assets/documents/34519, made available to FEMA by uploading the digital data to the MIP in accordance with the schedule outlined in Section 6.

Independent QA/QC: Hydraulics Data Capture

Responsible Mapping Partner: ISWS

Scope: The Independent QA/QC Mapping Partner shall perform an impartial review of the technical, scientific, and other information submitted by the CTP under Hydraulic Analysis to ensure that the data and modeling are consistent with FEMA standards, guidance, and standard engineering practice and are sufficient to revise the FIRM. FEMA may audit or assist in these activities if deemed to be necessary by the Regional Project Officer. This work shall include, at a minimum, the activities listed below.

Please note FEMA will also be performing periodic audits and overall study/project management to ensure study quality. The CTP will be responsible for addressing any and all comments resulting from independent QC, including resubmittal of deliverables as needed to pass technical review.

Review the submittal for technical and regulatory adequacy, completeness of required information, and supporting data and documentation. The technical review is to focus on the following:

- Use of acceptable model(s)
- Use of appropriate methodology(ies)
- Starting water-surface elevations
- Cross-section geometry
- Manning's "n" values and expansion/contraction coefficients
- Bridge and culvert modeling
- Ineffective and non-conveyance areas
- Flood discharges
- Regulatory floodway computation methods
- Tie-in to upstream and downstream non-revised Flood Profiles and floodways
- Agreement between the model, spatial data, work maps, Flood Profiles and Floodway Data Tables
- Calibration of model(s), where high-water marks are available
- Floodplain and floodway boundaries for the 1 percent and 0.2 percent annual chance events

- Verify that the data was submitted under the applicable GEOGRAPHIC FOOTPRINT folders in the MIP.
- Use the CHECK-2 or CHECK-RAS program, as appropriate, to flag potential problems and focus review efforts.
- Maintain records of all contacts, reviews, recommendations, and actions and make the data readily available to FEMA.
- Maintain an archive of all data submitted for hydraulic modeling review. (All supporting data must be retained for three years from the date a funding recipient submits its final expenditure report to FEMA, and once the study is effective all associated data should be submitted to the FEMA library).
- The reviewing Mapping Partner must document the results of the review in a memorandum or letter, send it to the Mapping Partner that performed the hydraulic analysis and post it to the MIP through the Independent QA/QC of Hydraulic Analyses task. The review document must present specific comments and may include any new calculations or model runs in support of the review.

<u>Standards</u>: All Independent QA/QC work shall be performed in accordance with the standards specified in Section 5 – Standards.

<u>Deliverables</u>: shall produce items listed in the Hydraulics Data Capture section within the most currently dated "Technical Reference: Data Capture" document, located at https://www.fema.gov/media-library/assets/documents/34519, made available to FEMA by uploading the digital data to the MIP in accordance with the schedule outlined in Section 6. Where paper documentation is required by state law for professional certifications, the CTP may submit the paper in addition to a scanned version of the paper for the digital record. Please coordinate with the regional and/or state representative to verify state reporting requirements.

- A Summary Report that describes the findings of the independent QA/QC review;
- Recommendations to resolve any problems that are identified during the independent QA/QC review.

Floodplain Mapping Data Capture

CTPs must complete this task in the MIP if a redelineation or digital conversion study is completed. A Floodplain Mapping MIP submittal is not required for revised data based on new or updated hydraulic or coastal analyses, as the required information shall be submitted with the Hydraulics Data Capture or Coastal Data Capture tasks. In addition, leveraged floodplain mapping data must be documented and submitted to the MIP for this task. Failure to submit this task when data should have been submitted is no longer an issue. The MIP redesign allows for tasks to be reopened so that data can be uploaded. If the CTP obtains floodplain mapping data that was not included in the MAS, the CTP shall contact the region to discuss the need to submit a change request to include the task and leverage as part of the award and request that this task be added to the MIP workflow.

Responsible Mapping Partner: ISWS.

The CTP shall perform floodplain mapping as described in Table 1.10: Summary of Floodplain Mapping. Floodplain Mapping activities include mapping and redelineation of the 1 percent and 0.2 percent annual chance event floodplains and regulatory floodways based upon updated topographic data. Floodplain Mapping may also include the digital conversion of non-revised floodplain area(s). Coastal flood sources shall also include mapping coastal static BFE zones as well as mapping the Limit of Moderate Wave Action (LiMWA).

The redelineation of floodplains utilizes updated topographic data to remap the floodplain boundary based upon effective water surface elevations. Per FEMA Standard ID #104, redelineation shall only be used when the terrain source data is better than effective, and the stream reach is classified as VALID in the CNMS database. Please review SID #104 listed in the last FEMA Policy for the most current language of the standard. Prior to redelineating effective floodplain boundaries, the current CNMS database should be reviewed to assess the appropriateness of this approach. The CTP shall use the effective FIS profile, floodway data table, FIRM/Flood Hazard Boundary Map (FHBM), and, as necessary, the effective hydraulic model, to delineate floodplain boundary on updated topography.

Per FEMA Standard ID #137, redelineation of coastal flood hazard areas requires the revision of the 1-percent-annual-chance SFHA boundary, the 0.2 percent-annual-chance floodplain boundary, and the primary frontal dune delineation. Please review SID #137 listed in the last FEMA Policy for the most current language of the standard.

Digital conversion of non-revised floodplain areas is the incorporation of the effective floodplain boundary, as-is, into the digital FIRM DB. This activity involves the georeferencing of effective paper maps and digitizing the effective floodplain and floodway boundaries. Digital conversions do not revise the effective Special Flood Hazard Area (SFHA).

Table 1.10 Summary of Floodplain Mapping

Study Area/Flooding Source	Method	Mapping Type (A/AE)	Miles	Topographic Data Source
Ohio River	Individual cross section datum conversion and redelineation	AE	28.2	

The CTP shall incorporate the results of all effective Letters of Map Change (LOMC) for all affected communities on the FIRM and provide to the appropriate PTS the required submittals for incorporation into the National Flood Hazard Layer (NFHL). Also, the CTP shall address all concerns or questions regarding Floodplain Mapping that are raised during the independent QA/QC review.

Per FEMA Standard ID #134, if the redelineation topographic data indicates that the effective hydraulic analyses are no longer valid, further actions must be coordinated with the FEMA Project Officer and the CNMS database must be updated. Please refer to SID #134 listed in the last FEMA Policy for the most current language of the standard. The CTP shall capture flood hazard engineering and/or mapping data quality issues encountered during this activity in the CNMS database for the area(s) of interest. These issues will be entered as “Requests” or “Needs” in the CNMS requests feature dataset based on the nature of the deficiency encountered. Detailed information on performing this task can be found in the relevant standards specified in Section 5 – Standards.

Standards: All floodplain mapping work shall be performed in accordance with the standards specified in Section 5 – Standards. The CTP will perform self-certification audits for the Floodplain Boundary Standards for all flood hazard areas.

Deliverables: The CTP shall produce items listed in the Floodplain Mapping Data Capture section within the most currently dated “Technical Reference: Data Capture” document, located at https://www.fema.gov/media-library/assets/documents/34519, made available to FEMA by uploading the digital data to the MIP in accordance with the schedule outlined in Section 6.

SECTION 2 – TECHNICAL AND ADMINISTRATIVE SUPPORT DATA SUBMITTAL

The Project Team members for this Flood Risk Project that have responsibilities for activities included in this MAS shall comply with the data submittal requirements summarized below and in appropriate guidance.

All supporting documentation for the activities in this MAS shall be submitted according to FEMA standards and requirements and will include a FEDD folder. Submittals must be made to the appropriate PTS for a review of required materials. The CTP will respond to requests from FEMA or its contractors for additional information and ensure that all required documents are included in the TSDN.

If any issues arise that could affect the completion of an activity within the proposed scope or budget, the CTP shall complete and submit to FEMA a Special Problem Report (SPR) as soon as possible after the issue is identified. The SPR describes the issue and proposes possible resolutions. For additional information on SPRs, consult the FEMA Regional Office.

Information supporting FEMA standards and requirements regarding the TSDN and FEDD file may be found in the "Technical Reference: Data Capture" document and other associated guidance documents.

Table 2.1: TSDN Section Mapping Activities

Mapping Activities	TSDN Section												
	General Documentation	Change Requests	Telephone Conversation	Meeting Minutes/ Reports	General Correspondence	Hydrologic Analyses	Engineering Analyses	Hydraulic Analyses	Key to Cross-Section Labeling	Key to Transect Labeling	Draft FIS Report	Mapping Information	Miscellaneous Reference
Survey Data Capture		X	X	X	X	X		X	X	X			X
Independent QA/QC: Survey Data Capture		X	X	X	X	X		X	X	X			X
New/Existing Topographic Data Capture		X	X	X	X							X	X
Terrain Data Capture		X	X	X	X							X	X
Base Map Data Capture		X	X	X	X	X		X	X	X	X	X	X
Hydrology Data Capture		X	X	X	X	X	X	X	X	X	X		X
Independent QA/QC: Hydrology Data Capture		X	X	X	X	X		X	X	X	X		X
Hydraulics Data Capture		X	X	X	X	X	X	X	X	X	X		X
Independent QA/QC: Hydraulics Data Capture		X	X	X	X	X		X	X	X	X		X

Mapping Activities	TSDN Section												
	General Documentation	Change Requests	Telephone Conversation	Meeting Minutes/ Reports	General Correspondence	Hydrologic Analyses	Engineering Analyses	Hydraulic Analyses	Key to Cross-Section Labeling	Key to Transect Labeling	Draft FIS Report	Mapping Information	Miscellaneous Reference
Floodplain Mapping Data Capture		X	X	X	X	X		X	X	X		X	X
Event Data Capture													

SECTION 3 – PERIOD OF PERFORMANCE

The mapping activities outlined in this MAS will be completed as specified in the Cooperative Agreement Funding Opportunity Announcement, Award Notice and/or Articles of Agreement. The Mapping Activities may be terminated at the option of FEMA or the CTP in accordance with the provisions of the Partnership Agreement dated September 9, 2013. If these mapping activities are terminated, all products produced to date must be submitted and updated into the MIP (if applicable) and the remaining funds, provided by FEMA for this MAS, from uncompleted activities will be returned to FEMA.

SECTION 4 – FUNDING/LEVERAGE (For CTP, OFA, and/or Community)

FEMA is providing funding, in the amount **$793,600** through a Cooperative Agreement to Illinois State Water Survey for the completion of this Flood Risk Project. During the discovery process, additional needs may be identified. Activities associated with any additional needs would be performed based on availability of additional funds. See Table 4.1.

Table 4.1: Contribution and Leverage

TASK, PARTNER, COST and SCHEDULE					
Project Task	FEMA Contribution	Partner Contribution (CTP Only)	Breakout Cost (if applicable)	% Partner (CTP Only) Leverage (of Total Project Task Cost)	Total Project Task Cost (FEMA + Partner)
Miscellaneous Tasks					
Project Management	$41,518.00	$$$	NA	%	$41,518.00
Project Risk ID & Mitigation	$32,901.00	$$$	NA	%	$32,901.00
Perform Community Engagement & Project Outreach	$14,035.00	$$$	NA	%	$14,035.00
Develop Regulatory Products & Datasets					
Survey Data Capture	$152,441.00	$$$	NA	%	$152,441.00
* Independent QA/QC: Survey Data	$30,488.00	$$$	NA	%	$30,488.00
Existing Topographic Data Capture	$16,450.00	$$$	NA	%	$16,450.00
Base Map Data Capture	$3,290.00	$$$	NA	%	$3,290.00
Hydrology Data Capture	$110,174.00	$$$	NA	%	$110,174.00
* Independent QA/QC: Hydrology Data	$11,017.00	$$$	NA	%	$11,017.00
Hydraulics Data Capture	$302,908.00	$$$	NA	%	$302,908.00
* Independent QA/QC: Hydraulics Data	$11,279.00	$$$	NA	%	$11,279.00
Floodplain Mapping: Redelineation Data Capture	$67,099.00	$$$	NA	%	$67,099.00
	$793,600	$0.00	$0.00	0%	$793,600.00

SECTION 5 – STANDARDS

The standards relevant to this MAS are presented in FEMA Policy 204-078-1 Standards for Flood Risk Analysis and Mapping, Revision 10, dated November 2019, located at http://www.fema.gov/media-library/assets/documents/35313. This Policy supersedes all previous standards included in the *Guidelines and Specifications for Flood Hazard Mapping Partners,* including all related appendices and Procedure Memorandums (PMs). Additional information and links to FEMA Technical References, guidance documents, templates and other resources may be accessed and downloaded at http://www.fema.gov/guidelines-and-standards-flood-risk-analysis-and-mapping. Revisions to the Policy memo are made on a regular basis. Some changes / updates are considered low impact, not requiring any scope, financial, process or technology changes to implement. CTPs should always check for the latest version of the policy memo to evaluate potential standards updates.

To facilitate the use of standards and related documents, users can access the FEMA *Guidelines and Standards Master Index* located here: www.fema.gov/media-library/assets/documents/94095. This index provides a cross-reference of documentation available for Flood Risk Projects, Letters of Map Change and related Risk MAP activities. The cross-referenced relationships are organized and accessible through linkages for standards, guidance, technical references and templates. The master index is updated in coordination with the FEMA Policy Memo noted above.

CTPs and their sub-awardees must comply with FEMA's Federal Regulations in Chapter 44 of the Code of Federal Regulations (CFR), specifically CFR Parts 65, 66 and 67, and the appropriate year CTP Notice of Funding Opportunity and Agreement Articles. CTPs shall also coordinate with their Regional office to determine additional standards that should be met.

SECTION 6 – SCHEDULE

To accurately document Risk MAP activities, there are two tools available for identifying mapping activities as they align to the Mapping Information Platform (MIP). The activities documented in this MAS shall be completed in accordance with either Table 6.1 Project Activities Schedule or the Mapping Information Platform (MIP) MAS/SOW Workbook.[2]

Please work with your FEMA Region when submitting the MIP MAS/SOW Workbook or any other form that is representative of the information in Table 6.1. Table 6.1, the MIP MAS/SOW Workbook or a similar table is required and needs to be filled out and documented in this MAS as part of the grant's submission process.

Table 6.1: Project Activities Schedule

Project Task	*Baseline Start Date*	*Baseline End Date*
Miscellaneous Tasks		
Project Management	09/01/2020	09/15/2024
Project Risk ID & Mitigation	09/15/2020	09/15/2024
Perform Community Engagement & Project Outreach	09/15/2020	09/15/2024
Develop Regulatory Products & Datasets		
Survey Data Capture	01/15/2021	01/15/2022
*** Independent QA/QC: Survey Data**	09/15/2021	03/15/2022
Existing Topographic Data Capture	03/15/2021	09/15/2021
Base Map Data Capture	03/15/2021	09/15/2021
Hydrology Data Capture	09/15/2021	09/15/2022
*** Independent QA/QC: Hydrology Data**	03/15/2022	09/15/2022
Hydraulics Data Capture	03/15/2022	09/15/2024
*** Independent QA/QC: Hydraulics Data**	03/15/2024	09/15/2024
Floodplain Mapping: Redelineation Data Capture	09/15/2023	09/15/2024

See R5_MAS_SOW worksheet tab "20-09_Massac"

The CTP will coordinate with FEMA, or its designee, to develop a baseline schedule for individual project activities. FEMA or its designee will utilize the individual project task schedule create the Flood Risk Project in the MIP and baseline the project activities with schedule and cost information within 30 days of

[2] You must sign into the CTP Collaboration Center to access this form. If you are not a registered user, please visit: https://www.surveymonkey.com/r/CTPCollaborationCenterRegistration

the funds being awarded and FEMA's approval of the final cost and schedule. The baseline schedule for individual project activities may be re-baselined in the MIP with approval from the FEMA Project Officer, and does not require a change to this MAS unless the overall project end date is modified.

SECTION 7 – CERTIFICATIONS

Data Capture

- **Data Capture:** Please refer to the current "Technical Reference: Data Capture" document, located at https://www.fema.gov/media-library/assets/documents/34519, for instructions on certifications. Generally, each Data Capture task includes certification forms and a project narrative. Mapping Partners should complete and submit only one Certification of Completeness and/or one Certification of Compliance form when the task is complete.

Perform Field Surveys and Develop Topographic Data

A Registered Professional Engineer or Licensed Land Surveyor shall provide an accuracy statement for field surveys and/or topographic data used and shall certify these data meet the accuracy statement provided. Data accuracy should be stated used the Federal Geographic Data Committee National Standards for Spatial Data Accuracy, but the American Society for Photogrammetry and Remote Sensing accuracy reporting standards are acceptable.

Prepare Base Map

- The CTP shall provide documentation that the digital base map can be used by FEMA and freely made available to the public. Please note that uploading Base Map data to the MIP does not constitute agreement that the digital base map can be used by FEMA. Documentation that the digital base map can be used by FEMA is still required.

Develop Hydrologic Data, Develop Hydraulic Data, Perform Coastal Analysis, and Perform Floodplain Mapping

- A Registered Professional Engineer shall certify hydrologic and hydraulic and coastal analyses and data in accordance with 44 CFR 65.6(f).
- Any levee systems to be accredited will be certified by the levee owner or other appropriate entity in accordance with 44 CFR 65.10.
- Certifications are required at the time the intermediate or final data is submitted.

SECTION 8 – TECHNICAL ASSISTANCE AND RESOURCES

Project Team members may obtain copies of FEMA-issued LOMCs, archived engineering backup data, and data collected as part of the CNMS process from FEMA and/or your Regional Project Officer.

General technical and programmatic information can be downloaded from the FEMA website at https://www.fema.gov/cooperating-technical-partners-program. Specific technical and programmatic support may be provided through FEMA and/or its contractor; such assistance should be requested through the FEMA Project Officer specified in Section 12 – Points of Contact.

Project Team members also may consult with the FEMA Regional Project Officer to request support in the areas of selection of data sources, digital data accuracy standards, assessment of vertical data accuracy, data collection methods or subcontractors, and GIS-based engineering and modeling training.

Please contact the region to obtain the most recent version of the Risk MAP timeline.

Assistance with the MIP may be requested at miphelp@RiskMAPcds.com.

SECTION 9 – CONTRACTORS

Contractor support may be used for all activities within this MAS, except staffing and mentoring, which must be completed by the CTP.

The CTP may use the services of a University of Illinois pre-approved firm providing land surveying services as a contractor for this MAS. The CTP shall ensure that the procurement for all contractors used for this Program Management Activity complies with the requirements of 2 CFR Part 200.

Guidance provided in this part includes, but is not limited to, contract administration and record keeping, notification requirements, review procedures, competition, methods of procurement, and cost and pricing analysis. The 2 CFR 200 documents may be viewed online at http://www.ecfr.gov/cgi-bin/text-idx?SID=cc011f4fb962e68cb0da4bc91e8fbb43&mc=true&node=pt2.1.200&rgn=div5. Additionally, contractors must not pose a conflict of interest issue.

SECTION 10 – REPORTING

Financial Reporting: Because funding has been provided to the CTP by FEMA, financial reporting requirements for the CTP will be in accordance with the terms of the Cooperative Agreement Funding Opportunity Announcement, Articles of Agreement or Award Notice for this MAS. The CTP shall also refer to 2 CFR Part 200.. ISWS through the University of Illinois shall provide financial reports to the FEMA Regional Project Officer and Assistance Officer in accordance with the terms of the signed Cooperative Agreement for this MAS.

Performance Reporting: Recipients are responsible for providing a signed performance report using the required list of information shown in the NOFO (or and old SF-PPR if you prefer) on a quarterly basis throughout the period of performance, including partial calendar quarters as well as for periods where no grant award activity occurs. The CTP shall refer to 2 CFR Part 200 to obtain minimum requirements for progress reporting. The Project Officer, as needed, may request additional information on progress.

The CTP may meet with FEMA and/or its contractor(s) as frequently as needed to review the progress of the project in addition to the quarterly financial and status submittals. These meetings may alternate between FEMA's Regional Office, the ISWS office, and conference calls, as necessary.

The CTP must report performance of the grant in conjunction with the progress reporting. The performance of the CTP is measured by the following criteria. Quantitative Targets for performance measures will be defined using the *FY2020 DHS NOFO for CTPs Appendix D Performance Measure References, "CTP Performance Measures Matrix* in conjunction with your FEMA PM and defined in Table 10.1.

Table 10.1: Performance Measures Targets

See CTP_Metrics_2020. xlxs for Summary of Performance Measures Targets

Output Measurement Options	Recorded Metric Unit/Scale	Massac County Data Development ISWS20-09
1. Net NVUE mileage	# of Miles (Initiated)	53 miles
[%] improvement gained between pre and post survey	Achieved / Not achieved	20%
1. Average scores from exit surveys. 2. [%] surveys completed by community officials/residents (# of completed surveys divided by # of public meeting participants who are community official/residents, not contractors)	Achieved / Not achieved	Avg. Score 3
1. Draft products require less than [#] revisions from FEMA or its contractors. 2. Risk MAP products will be provided to the appropriate entity within [# of days] of final product. 3. Final project data will be uploaded to the MIP within [# of days] of completion. 4. [# of] training participants.	Achieved / Not achieved	Measure #1 Goal 1 revision
[%] of key stakeholders attended ([#] of people invited divided by [#] of people attended)	Achieved / Not achieved	50%
1. [#] planned versus completed engagement(s) (% completed: completed engagements/plan engagements) with other project stakeholders 2. [#] of different partners invited/coordinated with in advance (diversity)	Achieved / Not achieved	1
0.92 to 1.08	SPI score (EV/planned value) CPI score (EV/actual cost)	0.92 to 1.08
1. Grant amendments that are not outside the CTPs span of control will be evaluated and documented. 2. Excessive CRs that are due to project management issues within the CTP's control will require additional discussion and possible development of a quality performance plan to identify a continued path to improved project and program management. 3. Reporting within specified timelines throughout the project.	# of CRs 0=no CRs 1=amendment needed outside of control 2=amendment needed within control 3=multiple amendments needed within control	Goal: 0
Monthly updates for all cases, including (%) of active tasks per	Achieved / Not achieved	Monthly MIP update
2. Ongoing and regular coordination with other state/local offices to discuss agency's prioritization process for Risk MAP projects as well as schedules, meetings, strategy development and stakeholder engagement.	Achieved / Not achieved	Monthly calls with FEMA & IDNR/OWR

The CTP shall communicate with communities throughout the life of each project. Continued engagement is necessary and appropriate and will build upon the relationships established or enhanced during Discovery and provide transparency into the Risk MAP process. This may occur through monthly or quarterly updates or project status calls with community leaders, project websites including updates at several milestones or along a specific timeline, or other methods.

SECTION 11 – PROJECT COORDINATION

Throughout the project, all members of the Project Team will coordinate, as necessary, to ensure the products meet the technical and format specifications required and contain accurate, up-to-date information. Coordination activities may include:

- Meetings, teleconferences, and/or video conferences with FEMA and other Project Team members will be conducted as per the communication plan.
- Telephone conversations with FEMA and other Project Team members on a monthly basis, and an ad hoc basis, as required.
- Updates to the MIP and other FEMA status information systems in accordance with FEMA standards and requirements.
- E-mail, facsimile transmissions, and letters, as required.

SECTION 12 – PROTECTION OF PRIVACY INFORMATION

To accomplish the tasks outlined in this scope, FEMA will provide the CTP access to Risk Analysis Management (RAM) which includes all PII collected in RAM as outlined in DHS/FEMA/PIA-045 Hazard Mitigation Planning and Flood Mapping Products and Services Support Systems. https://www.dhs.gov/publication/dhsfemapia-045-hazard-mitigation-planning-and-flood-mapping-products-and-services

The information sharing outlined in this scope authorized by the following System of Records Notice(s) and Routine Use(s): DHS/FEMA-014 Hazard Mitigation Planning and Flood Mapping Products and Services Records System of Records. https://www.federalregister.gov/documents/2017/10/25/2017-23205/privacy-act-of-1974-system-of-records

When necessary to accomplish a function related to this system of records, CTPs (term used to include any subrecipients and/or contractors used by the CTP for this work) must comply with the Privacy Act. Individuals provided information under this routine use are subject to the same Privacy Act requirements and limitations on disclosure as are applicable to DHS officers and employees.

The mapping program for the NFIP, implemented through Risk MAP, is established through the National Flood Insurance Act of 1968, as amended and the Biggert-Waters Flood Insurance Reform Act of 2012 (BW-12), as amended (42 U.S.C. 4001 et seq.). The mapping program is governed by the implementing regulations at Code of Federal Regulations - Title 44 (44 CFR) parts 59-72.

The CTP will limit access to the PII provided by FEMA under this contract only to the CTP's authorized personnel who need to know the information to accomplish the tasks outlined in this contract.

Definition. As used in this section, "personally identifiable information" means information that can be used to distinguish or trace an individual's identity, either alone or when combined with other information that is linked or linkable to a specific individual. (See Office of Management and Budget (OMB) Circular A-130, Managing Federal Information as a Strategic Resource).

A. The CTP will ensure that initial privacy training, and annual privacy training, thereafter, is completed by CTP employees, subrecipients and/or contractors and subcontractors who

i. Have access to a system of records.
ii. Create, collect, use, process, store, maintain, disseminate, disclose, dispose, or otherwise handle personally identifiable information on behalf of an agency; or
iii. Design, develop, maintain, or operate a system of records

B. FEMA will provide the link to external facing initial privacy training, and annual privacy training thereafter, to CTPs for the duration of this grant.
C. The CTP shall maintain and, upon request, provide documentation of completion of privacy training to FEMA.
D. The CTP shall not allow any employee access to a system of records, or permit any employee to create, collect, use, process, store, maintain, disseminate, disclose, dispose or otherwise handle personally identifiable information, or to design, develop, maintain, or operate a system of records unless the employee has completed privacy training, as required by this section.

The CTP will ensure no computer matching, as that term is defined in U.S.C. § 552a(o), will occur for the purpose of establishing or verifying eligibility or compliance as it relates to cash or in-kind assistance of payments under federal benefit programs.

If at any time during the term of this cooperative agreement any part of FEMA PII, in any form, that the CTP obtains from FEMA ceases to be required by the CTP for the performance of the scope, or upon termination of the cooperative agreement, whichever occurs first, the CTP shall, within fourteen (14) days thereafter, promptly notify FEMA and securely return PII to FEMA, or, at FEMA's written request destroy, un-install and/or remove all copies of such PII in the CTP's possession or control, and certify in writing to FEMA that such tasks have been completed.

SECTION 13 – POINTS OF CONTACT

The points of contact for this Flood Risk Project are John Wethington, the FEMA Regional Project Officer; Sally McConkey, the Project Manager for ISWS; or subsequent personnel of comparable experience who are appointed to fulfill these responsibilities. When necessary, any additional FEMA assistance should be requested through the FEMA Regional Project Officer.

Each party has caused this MAS to be executed by its duly authorized representative.

Sally McConkey

08/10/2020

Sally McConkey — Date
Project Manager

John Wethington — Date
Regional Project Officer
Federal Emergency Management Agency, Region 5

Julia McCarthy
Risk Analysis Branch Chief — Date
Federal Emergency Management Agency, Region 5

Mary Beth Caruso
Mitigation Division Director — Date
Federal Emergency Management Agency, Region 5